

SLM : Agriculture and Climate

Agriculture and Climate Advisory Dataset for Nigerian Smallholder Farmers

Dataset Name: NigeriaAgriClimate-SLM Corpus
Version: v1.0 - In Development
Date: August 2026
Team: Kulal
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Primary Language: English (Yoruba, Hausa, Igbo planned for future phases)
Task: Question and answer generation for smallholder farmer advisory
Model Target: Small Language Model (SLM) with FAISS-based semantic retrieval
Primary Datasets: FAO (Food and Agriculture Organization), CGIAR
Base Dataset: KisanVaani Agriculture Dataset
Geographic Scope: Sub-Saharan Africa
Status: In Development - Research Use Only

This dataset supports the training of a Small Language Model (SLM) built to give smallholder farmers in Nigeria practical guidance on agriculture and climate. The model uses a retrieval system built on FAISS to find relevant documents before generating a response, which keeps answers grounded in real source material. Data is collected in English first, with plans to extend to Yoruba, Hausa, and Igbo as more annotated material becomes available.

1. Dataset Composition

Primary Data Sources

Source	Description	Format	Volume
FAO	Food and Agriculture Organization publications covering crop management, climate adaptation, pest control, and food security guidance relevant to sub-Saharan Africa	PDF / text	~3,000 documents
CGIAR	Research outputs from CGIAR centres including IITA and ICRISAT covering Nigerian and West African agricultural systems, soil data, and climate-smart farming	PDF / text	~2,000 reports
KisanVaani	Large-scale Q&A dataset used as a base corpus to give the tokenizer a rich agricultural language foundation before Nigeria-specific fine-tuning	CSV / JSON	Thousands of Q&A pairs
Nigerian government	FMARD and ADP seasonal guides; NIMET historical rainfall and	PDF / CSV	~5,000 documents

	drought data by state and agroecological zone		
Farmer interviews	Community interviews with smallholder farmers in local languages, transcribed to English	Audio / text	500+ sessions
Synthetic data	Q&A pairs generated from verified source documents, reviewed by an agronomist before inclusion	JSON	Up to 20% of corpus

Record Schema

Each record in the processed dataset contains these fields:

Field	Description
id	Unique record identifier
source_type	FAO / CGIAR / government / farmer / KisanVaani / synthetic
language	en for English; yo / ha / ig added when translation is available
agroecological_zone	Sudan Savanna, Guinea Savanna, Derived Savanna, Rainforest Belt, or Mangrove
state	Nigerian state, or national for guidance that applies broadly
season	Dry, Wet, or Transition with approximate month range
crop_focus	Main crop referenced, e.g. maize, sorghum, cassava, groundnut
query	A farmer question written in plain English
response	An advisory answer grounded in verified source material
source_reference	Document ID or citation from the originating source
verified_by	Annotator ID and agronomist reviewer ID
date_added	Date the record was added, in ISO 8601 format

2. Data Collection Process

Existing and Published Sources

- FAO and CGIAR publications are accessed through their open-access portals, then cleaned and processed into Q&A pairs using the team's preprocessing pipeline.
- Nigerian government sources including FMARD and ADP guides are collected via official portals or direct institutional contact.
- KisanVaani is used as a base dataset to give the model a broad agricultural language foundation before Nigeria-specific fine-tuning.

Primary Collection - Farmer Interviews

Community sessions are conducted across at least four agroecological zones, supported by local agricultural extension officers. Sessions are held in the local language and transcribed into English for inclusion in the corpus.

- Format: Group interviews with 5 to 12 farmers, plus individual consultations.
- Facilitators: Bilingual field researchers with agricultural extension experience.
- Recording: Audio with written transcription. Participants can choose not to be recorded.
- Consent: Oral informed consent is obtained in the local language before every session.

Processing and Annotation Pipeline

Step	Description
1 - Data loading and EDA	Raw documents are ingested, cleaned, and explored. Column structure is defined based on source format. Duplicates and low-quality records are removed.
2 - Retrieval system	FAISS index is built over the cleaned corpus for semantic search. This replaces TF-IDF for better contextual matching.
3 - Agronomist review	A certified agronomist checks all Q&A pairs for factual accuracy before annotation begins.
4 - Annotation	Trained annotators label each record with zone, season, crop, source type, and intent category.
5 - Fine-tuning	The base model is fine-tuned on the curated Nigeria-specific corpus using the annotated Q&A pairs.
6 - Evaluation	Model outputs are evaluated for accuracy, relevance, and safety against a held-out test set reviewed by agronomists.

3. Representation and Bias

The table below lists known bias risks in the dataset, how each one was identified, and what steps are taken to reduce its impact.

Bias Type	Risk	Mitigation
Geographic	Southern and coastal zones may be underrepresented in FAO and government records that historically focus on the grain belt.	Minimum 15% representation target per agroecological zone. Gap analysis is run after each collection phase.
Gender	Women do 60 to 80% of smallholder farming in Nigeria but are underrepresented in formal agricultural records.	Community sessions actively recruit at least 50% female participants. Gender is logged per record.
Language register	Published advisory documents do not reflect how rural farmers actually ask questions.	Farmer interview data shapes the query language. Responses are written to a Grade 8 reading level or below.
Seasonal	FAO and CGIAR data may skew toward wet season guidance rather than dry season specifics.	Seasonal balance is checked across the dataset. Underrepresented seasons are prioritised in later collection rounds.

Crop diversity	Staple crops like maize, rice, and cassava are likely overrepresented compared to other locally important crops.	Minimum representation thresholds are set for at least 12 crop types common to Nigerian smallholders.
Base dataset domain	KisanVaani is India-focused. Farming practices may not match Nigerian context.	KisanVaani is used only as a language foundation. Nigeria-specific fine-tuning overrides region-specific advice.
Synthetic data	Model-generated Q&A pairs may contain errors or amplify existing biases.	Synthetic records are capped at 20% of the corpus. All synthetic records are flagged and reviewed before inclusion.

4. Labelling and Annotation

Annotators: Trained native English speakers with agricultural knowledge, recruited from Nigerian universities and extension programmes.

Label categories: Intent (pest / planting / climate / livestock / general), agroecological zone, season, crop focus, and language.

Agreement measure: Inter-annotator agreement measured using Cohen's Kappa. A score of 0.75 or above is required before a batch is accepted.

Disagreements: Discussed in weekly calibration sessions. A senior agronomist makes the final call.

Quality control: 10% of each annotator's weekly output is independently re-annotated and compared. Batches below threshold are fully re-done.

Tooling: Label Studio is used for annotation management. All annotation logs are kept for audit purposes.

5. Privacy and Consent

Personally Identifiable Information

- No farmer names, phone numbers, or farm locations are stored in the dataset.
- Location is recorded at community level only, using LGA, state, and agroecological zone.
- Annotator IDs are internal codes and are not included in any public release.

Consent

Written consent: Used where participants are literate. Forms are available in English and the relevant local language.

Oral consent: Used where participants prefer it. A field researcher witnesses and notes it in the session log.

Withdrawal: Participants can withdraw at any time before the dataset is compiled. Their records are fully removed.

Age: No one under 18 years old is included in primary data collection.

Indigenous Knowledge

Farming knowledge shared during community sessions belongs to the community, not to individual participants or the research team. Community leaders are briefed on the research purpose before any sessions begin. The dataset will not be sold or shared with third parties in ways that work against the community's interests. A plain-language summary of findings will be shared back with each participating community after the collection phase.

6. Intended and Out of Scope Use

Intended Uses

- Fine-tuning an SLM to answer agricultural and climate questions for Nigerian smallholder farmers.
- Academic research into natural language processing for low-resource African language settings.
- Testing AI-assisted extension service tools in sub-Saharan African farming contexts.
- Adapting the approach for other West African countries with similar farming systems, after proper regional review.

Out of Scope Uses

- Training a general-purpose large language model. This dataset is domain-specific and not suitable for that purpose.
- Deployment outside West Africa without independent localisation and agronomist validation.
- Commercial use without a data-sharing agreement with the originating institutions and communities.
- Replacing certified professional advice for individual high-stakes farming or veterinary decisions.
- Any use that involves tracking, profiling, or targeting individual farmers or communities.

7. Values Alignment

This table connects the values from the problem statement to specific decisions made in building the dataset.

Value	Why It Matters	What We Did
Equity and Inclusion	Over 70% of Nigeria's rural population works in agriculture, but current tools only work in English.	English is the starting language, with a clear plan to add Yoruba, Hausa, and Igbo. Gender and geographic balance targets are enforced.
Community Agency	Indigenous farming knowledge is left out of existing digital tools.	Farmers contribute through interview sessions. Community consent and data ownership are built into the process. Findings go back to communities.
Relevance	General tools do not account for local climate patterns or specific farming zones.	Every record includes agroecological zone, state, season, and crop. FAO and CGIAR data is filtered for Nigerian relevance.
Accessibility	Most rural farmers have limited connectivity and low exposure to formal written English.	Responses are written at Grade 8 reading level or below. The SLM is designed for low-bandwidth environments.
Transparency	Anyone using or evaluating this dataset should know where it came from.	Every record carries a source type, reference, and reviewer ID. This Data Card documents all design decisions including known gaps.
Avoiding Harm	Wrong advice can directly hurt a farmer's livelihood or food supply.	An agronomist reviews all records before inclusion. Synthetic data is capped and flagged. Out of scope uses are clearly defined.

8. Dataset Maintenance

Maintainers: Team Kulal in partnership with local agricultural institutions and extension programmes.

Updates: Updated twice a year, aligned with the Nigerian farming calendar: once before planting season and once before harvest.

Versioning: Semantic versioning is used. All versions are archived with a full changelog. Schema changes increment the major version number.

Error handling: Records with factual errors are flagged as deprecated rather than deleted, to keep the audit trail intact.

Feedback: Extension officers and NGO partners can submit corrections or flag harmful advice through a structured feedback form.

Review period: The dataset will be reviewed for continued relevance five years after release. Communities will be re-consulted before any major expansion.

9. Limitations and Open Questions

Known Limitations

- Multilingual data is scarce. Annotated agricultural text in Yoruba, Hausa, and Igbo is hard to find. The multilingual phase depends on finding the right community partners and funding.
- KisanVaani is India-focused. Care is taken not to let Indian farming context affect Nigerian advisory responses.
- NIMET climate data at LGA level may still miss micro-climate differences within a single community.
- Transcribing oral farmer knowledge into text can lose dialect variation, tone, and contextual meaning.
- This Data Card covers the dataset only. The SLM needs separate agronomic evaluation before field deployment.

Open Questions

- When indigenous knowledge from one community shapes advice given to another, how should that contribution be recognised?
 - How can farmers in low-connectivity areas report a wrong or harmful model response after deployment?
 - As climate patterns shift, how quickly should seasonal guidance in the dataset be updated to stay accurate?
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